

CS-4620 – Section 001

Compilers

Instructor:

Matthew Dwyer

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Office Hours: Wed. 2-3pm on via Zoom (link on canvas)

Teaching Assistants:

Soneya Binta Hossain

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Office Hours: Fri. 4-5pm via Zoom (link on canvas) or in-person (arrange via email)

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Office Hours: Fri. 11am-noon via Zoom (link on canvas) or in-person (arrange via email)

Course Description: This is an undergraduate course on compilers. The course has a broad focus on the major components of compilers and seeks to provide conceptual, algorithmic, and practical knowledge about these important software systems. The course goals are for students to (a) learn foundational concepts in compilation, (b) explore the realization of those concepts in practice in the “controlled” setting of an existing compiler, and (c) deepen and reinforce learning by extending that compiler in a semester long project.

The course blends the theory and algorithms with practical aspects of engineering compilers. Students are expected to have experience with software development from prior CS classes and to be familiar with C++. They will work, in pairs, on a project that will involve understanding a rather large code base written using modern C++ and extending it. The project will involve writing a substantial amount of code (upwards of 1000 SLOC across the semester), documenting that code, and providing test cases that demonstrate the correctness of that code. Students will use multiple tools to achieve this, but they will be provided with a model system to base their work on.

The project in this course is unique across undergraduate compiler courses in that it exposes students to modern software engineering tools, techniques and practices. It also involves the use one of the most widely used compiler infrastructures in use today – LLVM – which is at the heart of the clang compiler family. Consequently, students will gain significant practical

knowledge about software development that will make them better programmers and software developers.

Course Delivery: The course is offered through a combination of in-person and asynchronous delivery. Lectures on course topics are pre-recorded and available for student viewing asynchronously. Weekly in-person sessions will provide ample opportunity for students to (1) ask questions, discuss answers with each other, and with the instructor, and (2) report on their project progress.

In-person sessions will be on Tues. and Thurs. from 3:30-4:45pm in room E316 of Thornton Hall. Typically, we will use the Tuesday meetings for (1) and the Thursday meetings for (2), but in the beginning of the course we will use the in-person meetings a bit differently, e.g., traditional lecture, demonstrations, etc.

We have found that office hours for this class work much better over zoom. This allows us to easily share screens and see exactly what problems you are having (as opposed to huddling around your laptop). So, for pedagogical reasons we will use zoom for office hours for the most part (some TAs may choose to offer in-person office hours).

Prerequisite(s): CS 2150 is required and CS 3330 is recommended.

Credit hours: 3

Text(s): No textbook is required. Collab site provides links to lecture notes and readings.

Assignments and Grading Percentages

- Project: 50% (5 deliverables including final project, each worth 10 points). Projects solved by pairs of students and you may pick your teammate.
- Weekly Quizzes: 25% (11 15 minute in-class quizzes, each worth 3 points, max points capped at 25 out of 33). Quizzes are given on canvas so bring a laptop to class on Tuesdays. We provide more quiz points than you can count for your grade. If you miss a Tuesday class and the quiz, there is no need to contact me. We will just total up all of the quizzes you do take and cap the points at 25. There are no makeup quizzes.
- 2 Exams: 12.5% each (one in Oct. and one in Dec. on first and second half of course content, respectively). No final exam for the course.

Letter Grade Cutoffs

A+ [98,-], A [93,98), A- [90,93), B+ [87,90), B [83,87), B- [80,83), C+ [77,80),
C [73,77), C- [70,73), D+ [67,70), D [63,67), D- [60,63), F [-,60)

Course Policies

- Students must comply with the University Honor Code (see below)

- The course project will be completed by students working in pairs, but discussion of solutions and sharing of information across pairs of students is not permitted and is considered an Honor Code violation.
- Late assignments will not receive a grade.
- The instructor and TAs will answer questions up to 48 hours before the project due dates. Projects are due at 5pm on Fridays, so any questions prior to 5pm on Wednesdays will be answered and any questions after that will not.
- You may use generative models, like ChatGPT3, on quizzes. If you do you must include the sequence of prompts used to produce your solution along with your solution.
- You may use generative models, like ChatCPT3, on projects. If you do you must include a 1000-word essay describing how you used the model, what benefit it provided, and where it was not as helpful as you had hoped.

Academic Integrity

The School of Engineering and Applied Science relies upon and cherishes its community of trust. We firmly endorse, uphold, and embrace the University's Honor principle that students will not lie, cheat, or steal, nor shall they tolerate those who do. We recognize that even one honor infraction can destroy an exemplary reputation that has taken years to build. Acting in a manner consistent with the principles of honor will benefit every member of the community both while enrolled in the Engineering School and in the future.

Students are expected to be familiar with the university honor code, including the section on academic fraud. Each assignment will describe allowed collaborations, and deviations from these will be considered Honor violations. If you are in doubt regarding the requirements, please consult with me before you complete any requirement of this course. Send, receiving or otherwise copying electronic files that are part of course assignments are not allowed collaborations (except for those explicitly allowed in assignment instructions).

Assignments where honor infractions or prohibited collaborations occur will receive a zero grade for that entire assignment or exam, as well as a full letter-grade penalty on the course grade. Such infractions will also be submitted to the Honor Committee if that is appropriate. Students who have had prohibited collaborations may not be allowed to work with partners on remaining homeworks.

SDAC and Other Special Circumstances

The University of Virginia strives to provide accessibility to all students. If you require an accommodation to fully access this course, please contact the Student Disability Access Center (SDAC) at 434-243-5180 or sdac@virginia.edu. If you are unsure if you require an accommodation, or to learn more about their services, you may contact the SDAC at the number above or by visiting their website at [URL](#).

If you have been identified as an SDAC student, please let the Center know you are taking this class. If you suspect you should be an SDAC student, please schedule an appointment with them for an evaluation. Students who need academic accommodations should see me and contact the SDAC. All academic accommodations must be arranged through the SDAC.

If you have other special circumstances (athletics, other university-related activities, etc.) please contact instructor and/or TA as soon as you know these may affect you in class.

Religious Accommodations

It is the University's long-standing policy and practice to reasonably accommodate students so that they do not experience an adverse academic consequence when sincerely held religious beliefs or observances conflict with academic requirements. Students who wish to request academic accommodation for a religious observance should submit their request in writing directly to me by email as far in advance as possible. Students and instructors who have questions or concerns about academic accommodations for religious observance or religious beliefs may contact the University's Office for Equal Opportunity and Civil Rights (EOCR) at UVAEOCR@virginia.edu or 434-924-3200.

Accommodations do not relieve you of the responsibility for completion of any part of the coursework missed as the result of a religious observance.

Statement on Violence

The University of Virginia is dedicated to providing a safe and equitable learning environment for all students. To that end, it is vital that you know two values that I and the University hold as critically important:

Power-based personal violence will not be tolerated. Everyone has a responsibility to do their part to maintain a safe community on Grounds. If you or someone you know has been affected by power-based personal violence, more information can be found on the UVA Sexual Violence website that describes reporting options and resources available
- www.virginia.edu/sexualviolence.

As your professor and as a person, know that I care about you and your well-being and stand ready to provide support and resources as I can. As a faculty member, I am a responsible employee, which means that I am required by University policy and federal law to report what you tell me to the University's Title IX Coordinator. The Title IX Coordinator's job is to ensure that the reporting student receives the resources and support that they need, while also reviewing the information presented to determine whether further action is necessary to ensure survivor safety and the safety of the University community. If you would rather keep this information confidential, there are Confidential Employees you can talk to on Grounds (See <http://www.virginia.edu/justreportit/>). The worst possible situation would be for you or your friend to remain silent when there are so many here willing and able to help.

This syllabus

This syllabus is to be considered a reference document that can and will be adjusted through the course of the semester to address changing needs. This syllabus can be changed at any time without notification. It is up to the student to monitor this page for any changes. Final authority on any decision in this course rests with the professor, not with this document